

AI Data Science Workbench - California Housing Project Report

Governed, reproducible median-house-value regression

Generated 2026-07-20T21:37:17+00:00 - target: median_house_value

1. Project overview

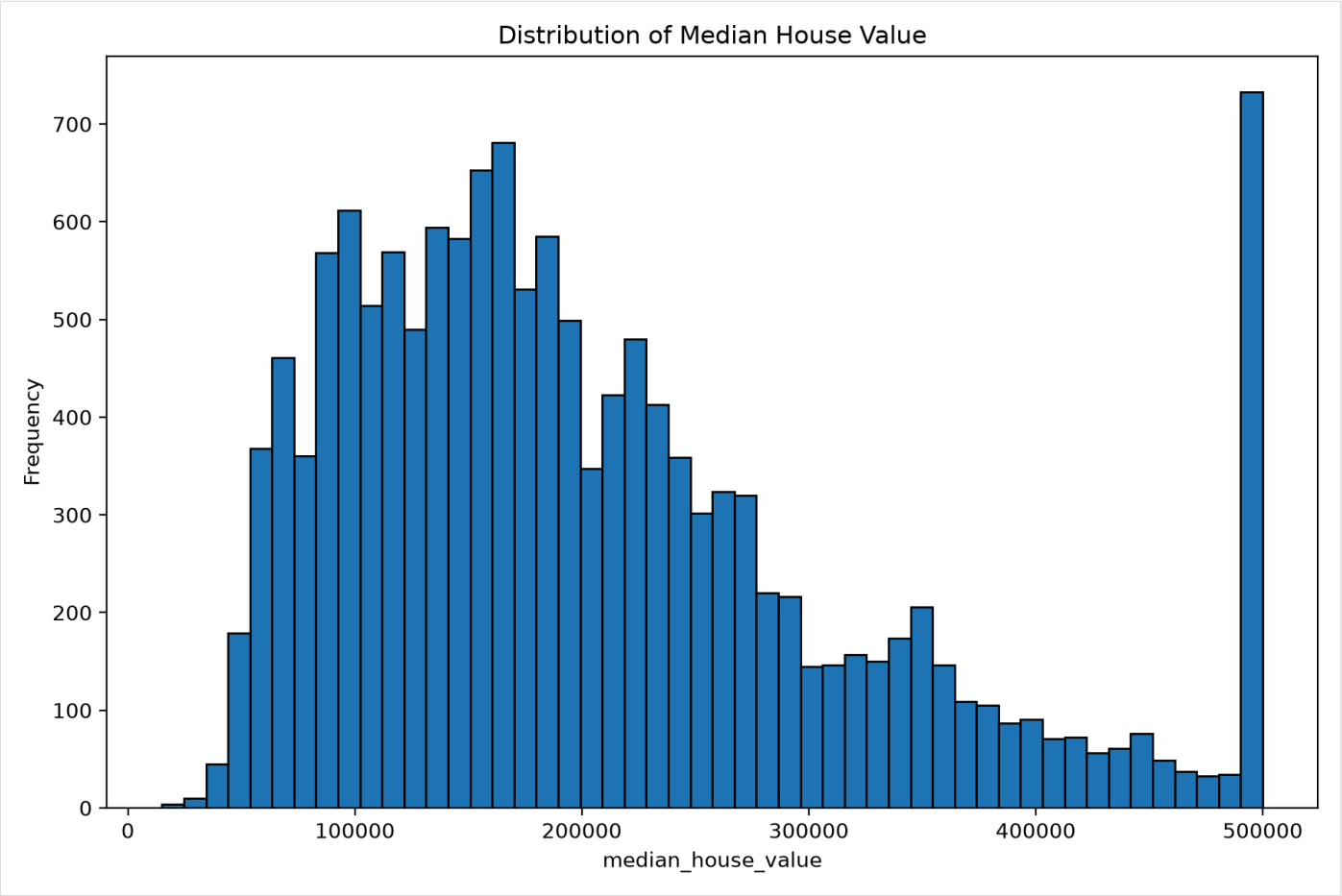
Linear family (LinearRegression, Ridge, Lasso); catalogue extensible.

2. Data validation summary

Status: Validation status: PASSED WITH WARNINGS

observations	columns	validation_errors	validation_warnings	duplicate_rows	total_missing_values	status
20640	10	0	1	0	207	PASSED WITH WARNINGS

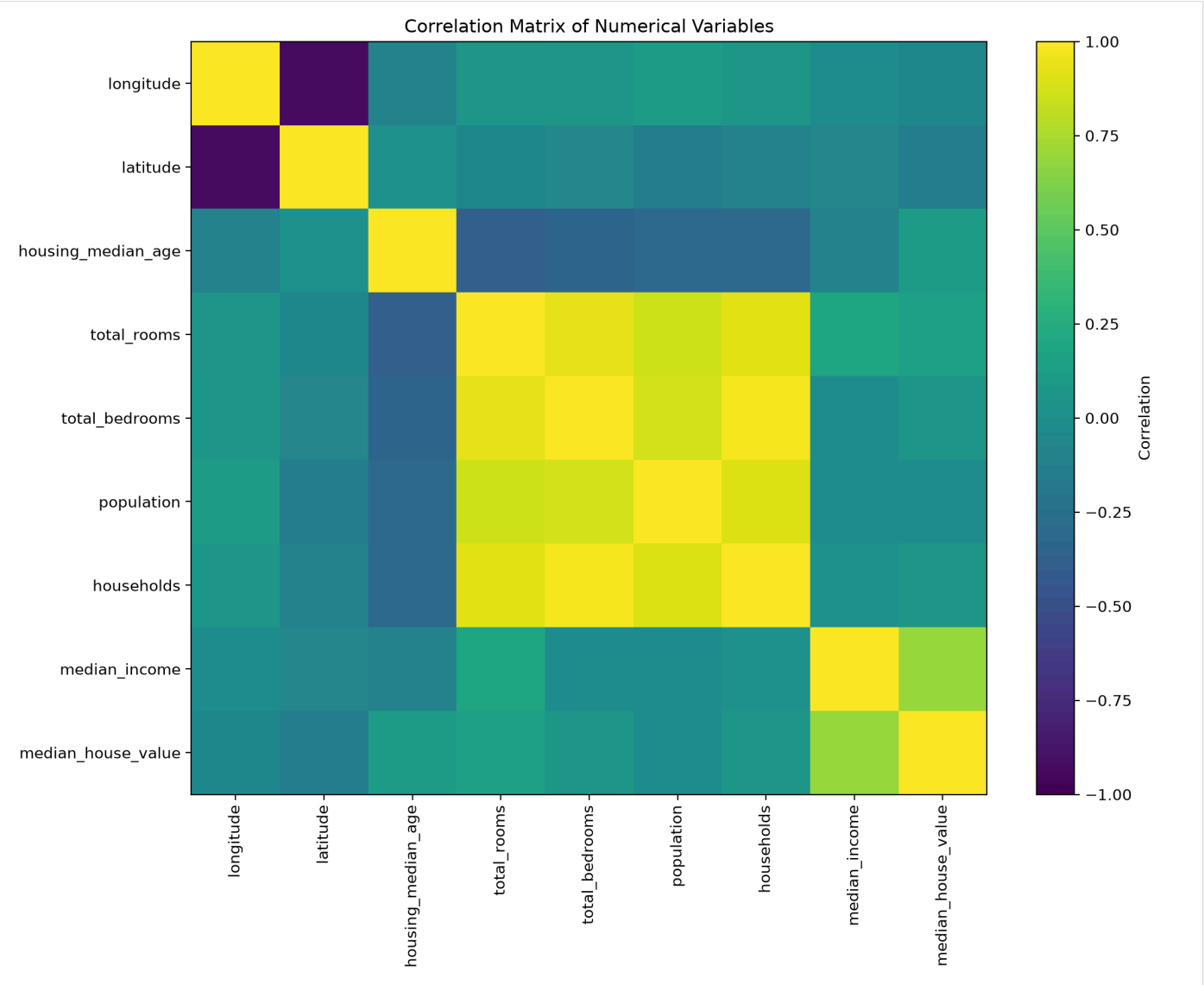
3. Key EDA figures



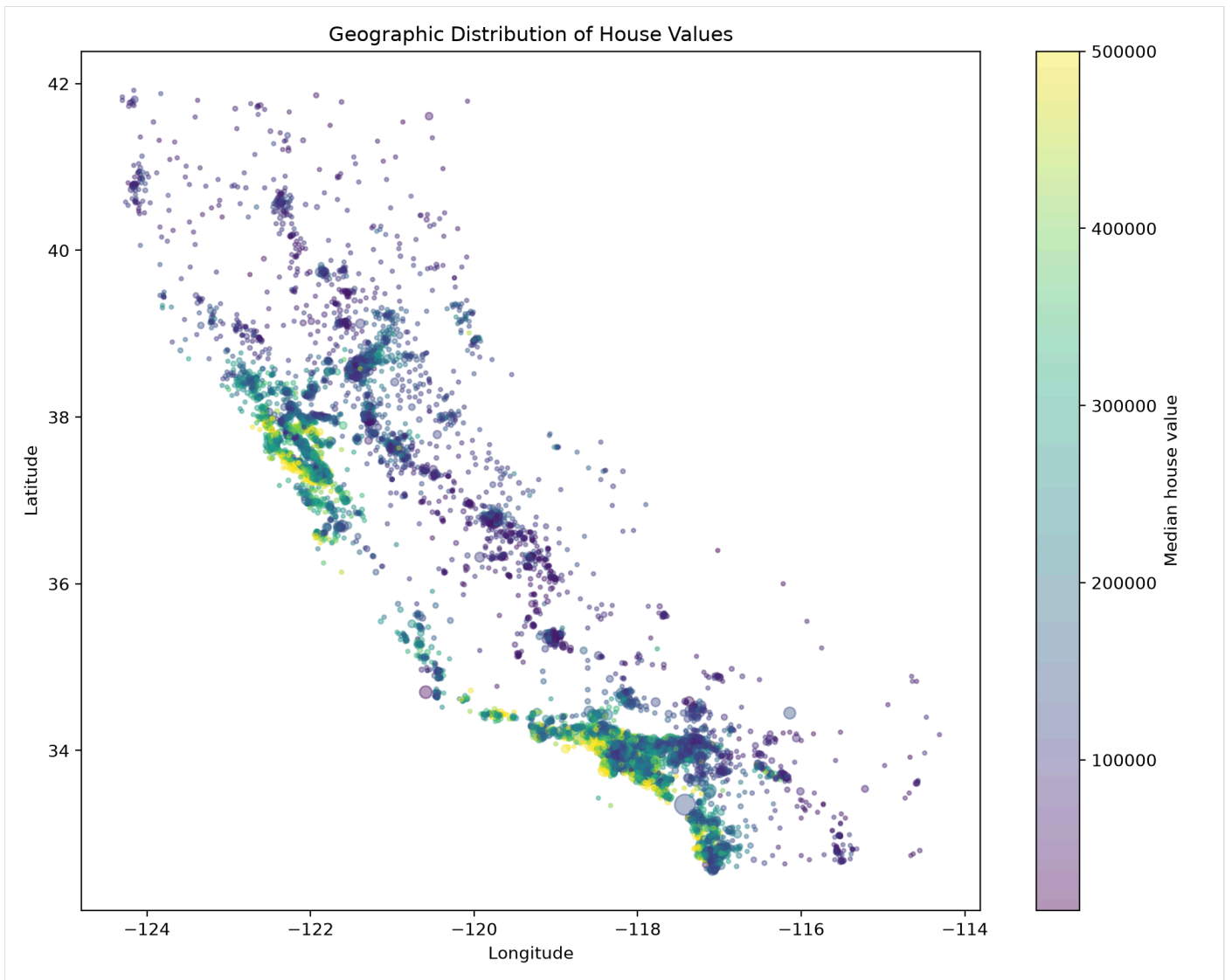
Distribution of median house value (the target).



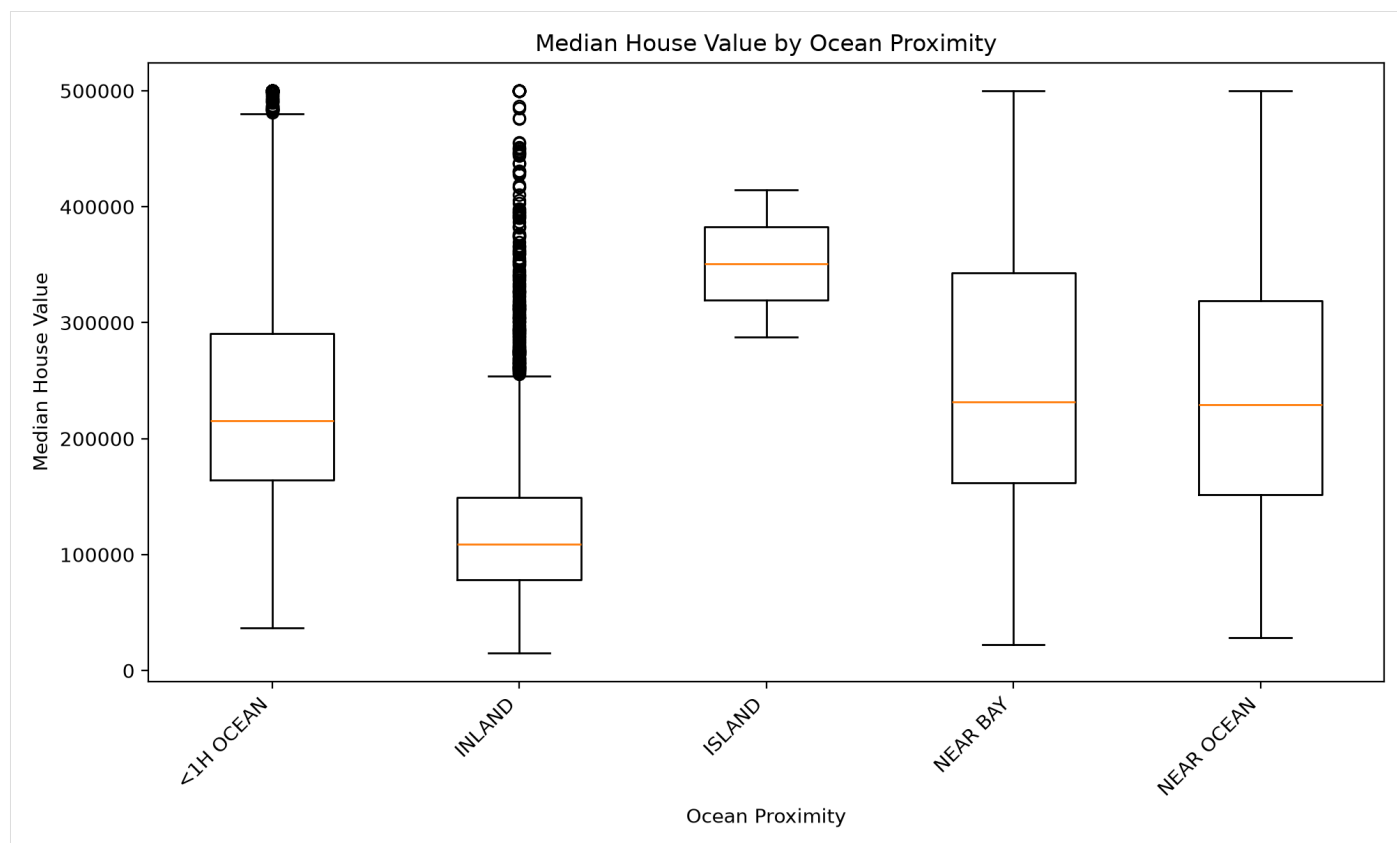
Median income vs house value - the strongest single relationship.



Correlation matrix across all numeric predictors.



Geographic distribution of house value (coastal premium is visible).



House value by ocean-proximity category.

4. Feature plan and approvals

16 features proposed; 10 approved for modelling.

Feature	Formula	Business meaning	Automatic	Approved
rooms_per_household	$\text{total_rooms} / \text{households}$	Average housing-space availability per household.	True	True
bedrooms_per_room	$\text{total_bedrooms} / \text{total_rooms}$	Bedroom share within the housing stock.	True	True
population_per_household	$\text{population} / \text{households}$	Average household occupancy.	True	True
bedrooms_per_household	$\text{total_bedrooms} / \text{households}$	Average bedroom availability per household.	True	True
rooms_per_person	$\text{total_rooms} / \text{population}$	Housing-space availability per person.	True	True
bedrooms_per_person	$\text{total_bedrooms} / \text{population}$	Bedroom availability per person.	True	True
coastal_access_group	Business-approved grouping of proximity categories	Simplified market-access / coastal-amenity segment.	False	False
log_total_rooms	$\log_{1p}(\text{total_rooms})$	Compresses the right-skewed scale of total_rooms.	True	True
log_total_bedrooms	$\log_{1p}(\text{total_bedrooms})$	Compresses the right-skewed scale of total_bedrooms.	True	True
log_population	$\log_{1p}(\text{population})$	Compresses the right-skewed scale of population.	True	True
log_households	$\log_{1p}(\text{households})$	Compresses the right-skewed scale of households.	True	True

Feature	Formula	Business meaning	Automatic	Approved
longitude_latitude_interaction	longitude * latitude	Joint geographic effect.	False	False
income_age_interaction	median_income * housing_median_age	Income effect may vary with housing age.	False	False
standardized_numeric_predictors	StandardScaler fitted on training data	Places numerical predictors on comparable scales.	False	False
one_hot_categorical_predictors	OneHotEncoder(handle_unknown='ignore') fitted on training data	Represents categorical differences without imposing order.	False	False
centered_numeric_predictors	x - training mean	Makes the intercept and interaction terms interpretable.	False	False

5. Model proposal, approval and comparison

Recommended: LinearRegression, Ridge, Lasso. Approved: LinearRegression, Ridge, Lasso.

Validation comparison (selection by RMSE)

Rank	Model	Version	Val RMSE	Val MAE	Val R2
1	Ridge	v001	64787.47785361487	46246.33782090241	0.680133757870843
2	Ridge	v002	64787.47785361487	46246.33782090241	0.680133757870843
3	Lasso	v002	64887.43494233497	46300.76369616628	0.6791459878872211
4	Lasso	v001	64887.43494233497	46300.76369616628	0.6791459878872211
5	LinearRegression	v002	64887.43759422965	46300.76391775423	0.6791459616611555
6	LinearRegression	v001	64887.43759422965	46300.76391775423	0.6791459616611555

6. Champion evaluation (untouched test set)

Champion: **Ridge v001**

66,680.81

Test RMSE

48,731.42

Test MAE

0.67

Test R2

64,787.48

Val RMSE

Three linear candidates were compared on the untouched validation split using RMSE as the selection metric.

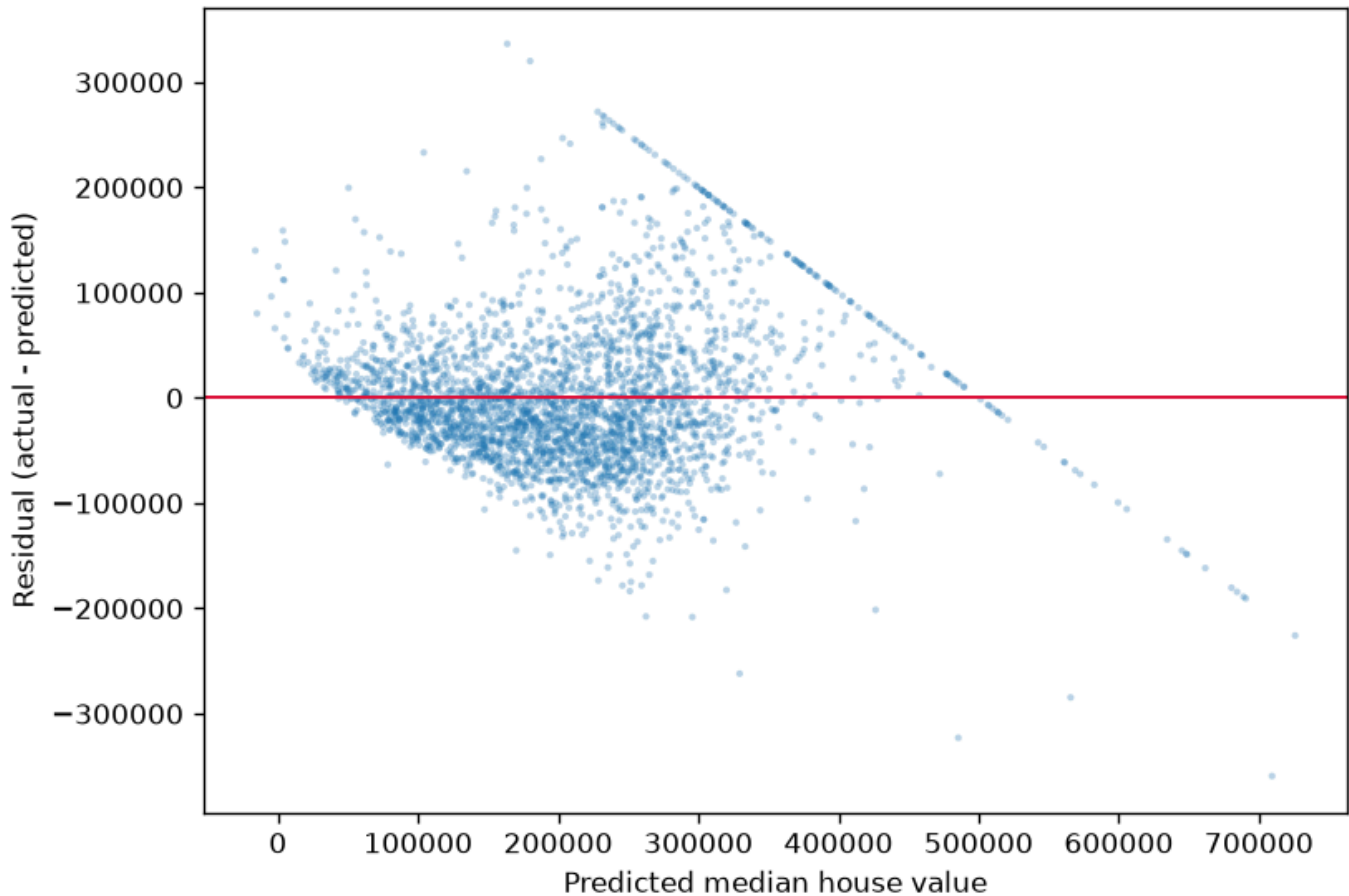
- Ridge (v001): validation RMSE 64,787, MAE 46,246, R2 0.680.
- Ridge (v002): validation RMSE 64,787, MAE 46,246, R2 0.680.
- Lasso (v002): validation RMSE 64,887, MAE 46,301, R2 0.679.
- Lasso (v001): validation RMSE 64,887, MAE 46,301, R2 0.679.
- LinearRegression (v002): validation RMSE 64,887, MAE 46,301, R2 0.679.
- LinearRegression (v001): validation RMSE 64,887, MAE 46,301, R2 0.679.

Ridge (v001) had the lowest validation RMSE and was promoted to champion. It beat the next-best model (Ridge) by 0 RMSE.

The champion was then evaluated once on the held-out test split: RMSE 66,681, MAE 48,731, R2 0.667.

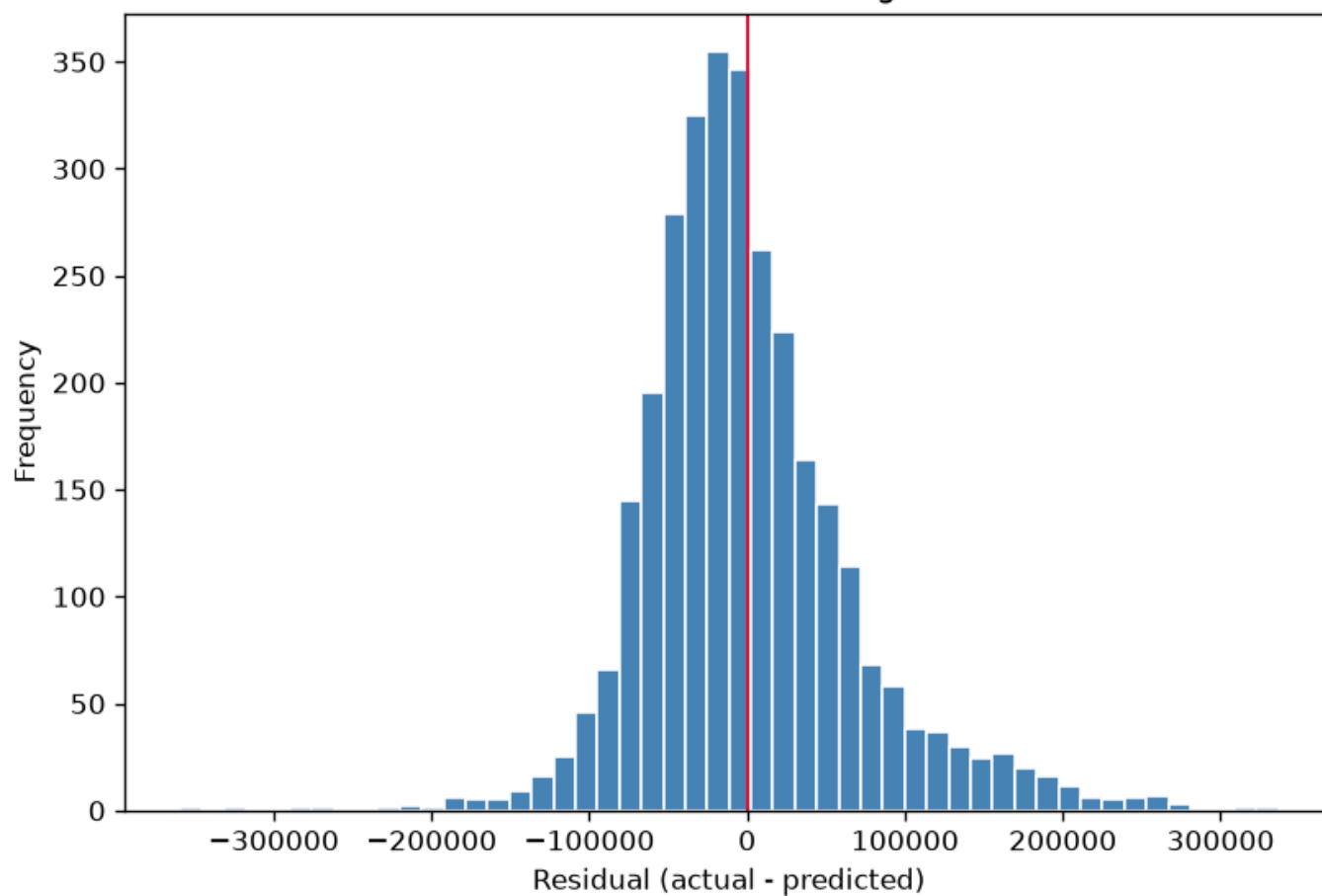
The test split was used only for this single final measurement, never for tuning or selection.

Residuals vs Predicted - Ridge v001



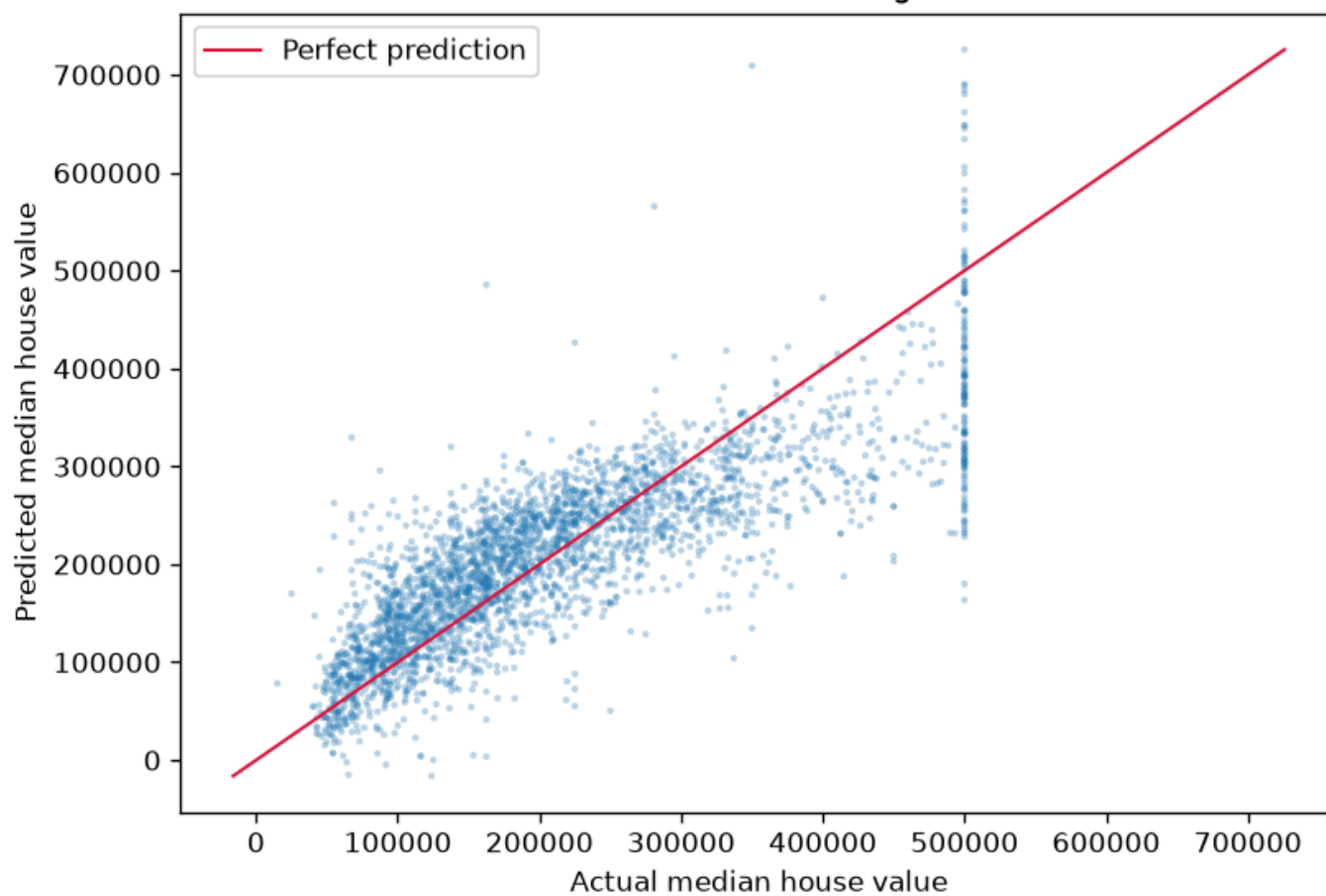
Residuals Vs Predicted

Residual Distribution - Ridge v001



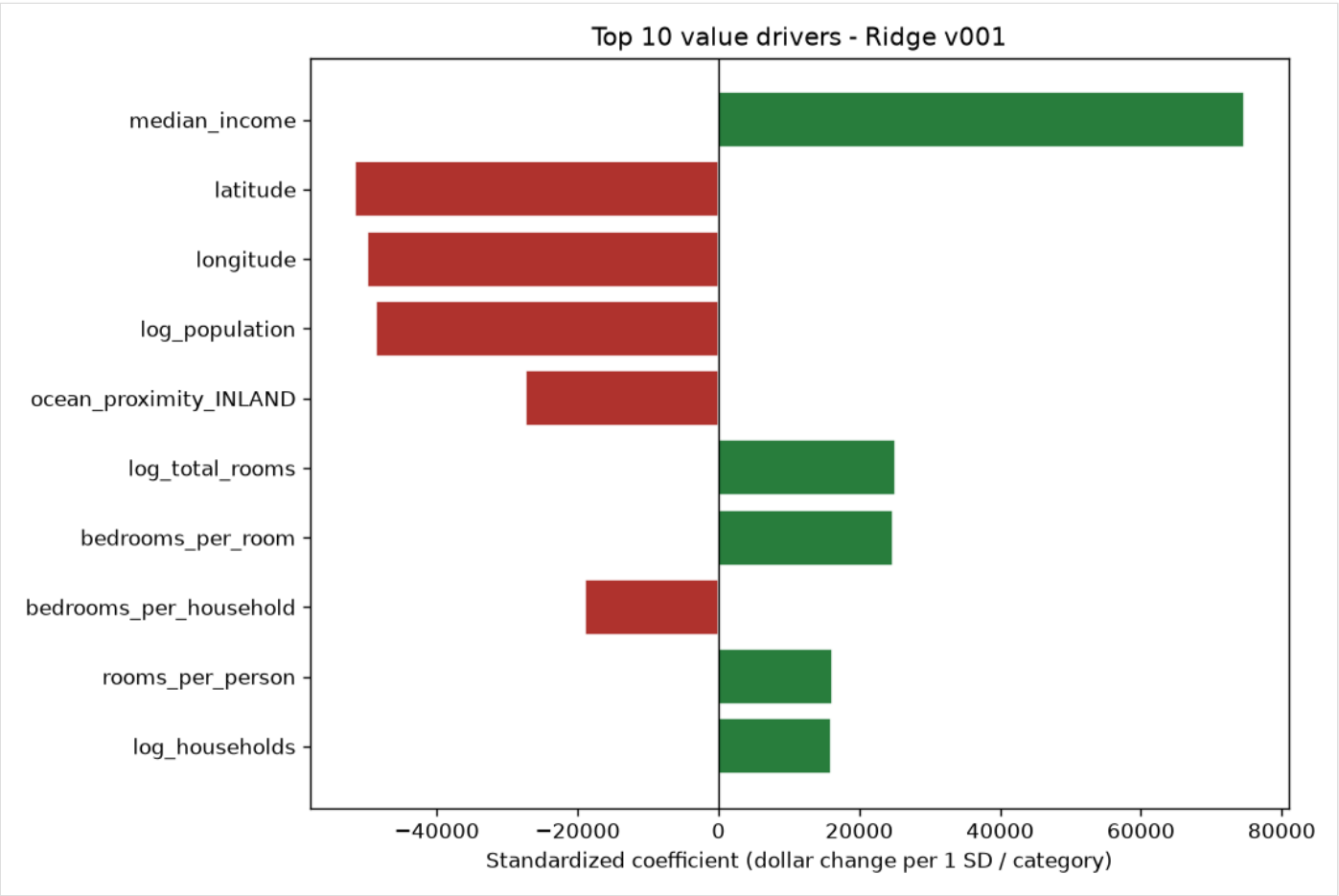
Residual Histogram

Predicted vs Actual - Ridge v001



Predicted Vs Actual

7. Coefficient interpretation



Top standardized value drivers.

Rank	Feature	Coefficient	Direction	Plain-English meaning
1	median_income	74630.48725742806	increases value	District median household income (the dominant driver of value).
2	latitude	-51647.20988728948	decreases value	North-south location (proxy for region within California).
3	longitude	-49816.15426551952	decreases value	East-west location (proxy for coastal/inland position).
4	log_population	-48747.34937611208	decreases value	Scale of population (skew-compressed).
5	ocean_proximity_INLAND	-27387.838468912894	decreases value	Home is inland (typically the lowest-value segment).
6	log_total_rooms	25133.808448278483	increases value	Scale of the housing stock (skew-compressed total rooms).
7	bedrooms_per_room	24798.11059053798	increases value	Bedroom share of rooms; higher values flag smaller/denser units.
8	bedrooms_per_household	-18899.58886329999	decreases value	Average bedrooms available per household.
9	rooms_per_person	16238.498548451193	increases value	Living space per resident.
10	log_households	16046.009894747682	increases value	Scale of households (skew-compressed).
11	housing_median_age	14321.680110714207	increases value	Typical age of the housing stock in the district.
12	households	14212.14612530292	increases value	Number of households in the census block.
13	ocean_proximity_<1H OCEAN	11595.234671068132	increases value	Home is within an hour of the ocean (premium coastal access).
14	ocean_proximity_NEAR OCEAN	9432.648100877024	increases value	Home is near the ocean (coastal amenity premium).
15	log_total_bedrooms	8753.175147859178	increases value	Scale of bedrooms (skew-compressed total bedrooms).

8. Governance summary

21

Audit events

3

Lineage nodes

5

Recorded decisions

Audit events by type

Event type	Count
model_proposal_created	3
feature_approval_granted	2
model_approval_granted	2
model_training_run	8
model_evaluation_run	2
final_model_selected	2
explainability_generated	2

Lineage stages

model_training -> model_evaluation -> explainability

9. Managerial recommendations

Executive Briefing: What Drives California House Prices

Prepared 2026-07-20T21:37:17+00:00 - champion model: Ridge v001

Business question

Which measurable characteristics of a California census block most affect its median home value, and can we predict that value reliably enough to support portfolio, pricing, and market-entry decisions?

The data

The analysis uses the California census housing dataset: roughly 20,600 census blocks, each described by location, housing age, room and bedroom counts, population, households, median income, coastal proximity, and the target, **median_house_value**. The data was split into training, validation, and an untouched test set; every statistic used to clean and scale the data was learned from the training portion only, so the reported accuracy is an honest estimate of performance on unseen blocks.

Method, in one paragraph

We fit a family of transparent linear models in which every input is placed on a common scale, so each model produces one easily-read number per factor: how many dollars the predicted value moves when that factor increases by a typical amount. We compared the candidates on the validation set, selected the best performer (**Ridge**), and measured it once on the held-out test set. No opaque black-box techniques were used, so every prediction can be explained to a stakeholder in plain business terms.

How well it predicts

- Validation error (RMSE): about \$64,787
- Validation R-squared: 0.680 (share of value variation explained)
- Baseline predicted value (intercept): about \$209,064

Top 5 value drivers

1. **median_income** - District median household income (the dominant driver of value). Each one-standard-deviation move raises predicted median value by about \$74,630.
2. **latitude** - North-south location (proxy for region within California). Each one-standard-deviation move lowers predicted median value by about \$51,647.
3. **longitude** - East-west location (proxy for coastal/inland position). Each one-standard-deviation move lowers predicted median value by about \$49,816.
4. **log_population** - Scale of population (skew-compressed). Each one-standard-deviation move lowers predicted median value by about \$48,747.
5. **ocean_proximity_INLAND** - Home is inland (typically the lowest-value segment). Each one-standard-deviation move lowers predicted median value by about \$27,388.

Important caveats

- **Multicollinearity among room/household counts.** Total rooms, bedrooms, population, and households move together; their individual coefficients should be read as a group, not as independent levers.
- **Coastal nonlinearity.** The value premium for coastal proximity is not a straight line; the linear model approximates it with category indicators and will understate sharp coastal effects.
- **Census-block aggregation.** Every figure is a block-level average, not an individual house; conclusions apply to neighbourhoods, not single properties.
- **Target cap.** Median values are capped at ~\$500,001 in the source census data, so the model cannot distinguish the most expensive blocks from one another and will under-predict at the very top of the market.

Recommendations

1. Treat **median income** and **coastal proximity** as the primary levers when screening markets; they dominate predicted value.
2. Use the model for **relative** ranking of neighbourhoods rather than precise single-home valuation, given block-level aggregation and the price cap.
3. For the highest-value coastal segments, supplement the linear model with local expertise, since the linear form understates coastal premiums.
4. Revisit the model if new data lifts the value cap or adds property-level detail, which would materially improve top-of-market accuracy.